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Evolution of Chemoautotrophic Endosymbioses in Bivalves

Bivalve–bacteria chemosymbioses are phylogenetically diverse but morphologically similar

Daniel L. Distel

Hydrothermal vents conjure the very images that are associated with primal life on a newly formed Earth: massive stone chimneys rising from a dark sea-floor, plumes of sulfurous smoke billowing from their peaks, and, all around, strange creatures feeding silently in the shimmering geothermal heat. It seems only fitting that the hydrothermal vent environment is posited to be a refuge for species derived from primitive and ancient forms (Newman 1985, Tunnicliffe 1992). However, one of the most unusual biological phenomena discovered at the vents now appears to have a long history spent in less exotic environments far removed from hydrothermal vents.

This phenomenon is chemoautotrophic endosymbiosis, a nutritional strategy found in an increasing number of marine invertebrates. With the aid of bacterial endosymbionts, these “chemosymbiotic” animals subsist on nutrients and energy sources, including hydrogen sulfide and carbon dioxide, that are normally unavailable to animals. Similar strategies have been observed in at least five animal phyla, including bivalve and gastropod molluscs and vestimentiferan, pogonophoran, annelid, and nematode worms. It is among the bivalves, however, that

The diversity and antiquity of chemo-symbiotic bivalve and bacterial lineages suggest that chemosymbiosis may have played an important role in the early diversification of the Bivalvia

this strategy is most widespread, both phylogenetically and geographically. In this article, I examine where, when, and how these symbioses became established in this important and diverse invertebrate group.

In 1977, near the Galápagos Islands, geologists got their first glimpse of the deep-sea hot springs known as hydrothermal vents. The existence of these volcanic features was anticipated by theories of plate tectonics and sea-floor spreading; however, the discovery of lush biological communities surrounding the vents (Lonsdale 1977) was completely unexpected. These diverse and densely populated communities were remarkable for a number of reasons. First, they stand in sharp contrast to the long-held perception of the deep-sea floor as a uniformly spare and desertlike environment. Second, they are populated largely by species and higher taxa that are completely new to science (Newman 1985, Tunnicliffe 1992).

And third, the basis of primary productivity in the vent communities is not photosynthesis, as is the case for all previously known biological communities, but bacterial chemosynthesis—rather than sunlight, these communities use geothermal energy stored in the form of thermogenic hydrogen sulfide and other reduced inorganic compounds. Even more remarkable, however, was the discovery that many of the bacteria responsible for this primary production exist as chemoautotrophic endosymbionts that live intracellularly within specialized organs in many vent-associated invertebrates (Cavanaugh et al. 1981, Felbeck et al. 1981).

Although fundamentally new to science, chemoautotrophic endosymbiosis is a nutritional strategy that loosely parallels photosynthesis in higher plants. In plants, photosynthesis occurs in chloroplasts, organelles that are now believed to have evolved from endosymbiotic photosynthetic bacteria (cyanobacteria). Both chemoautotrophic endosymbionts and photoautotrophic chloroplasts use ATP as an energy source to drive fixation of carbon dioxide into carbohydrates via the Calvin–Benson Cycle. These carbohydrates subsequently serve as the primary nutritional carbon source for their respective hosts. The two “symbioses” differ, however, in the way in which ATP is generated: Chloroplasts use light energy to synthesize ATP, whereas chemoautotrophic endosymbionts harness chemical energy stored in reduced sulfur com-

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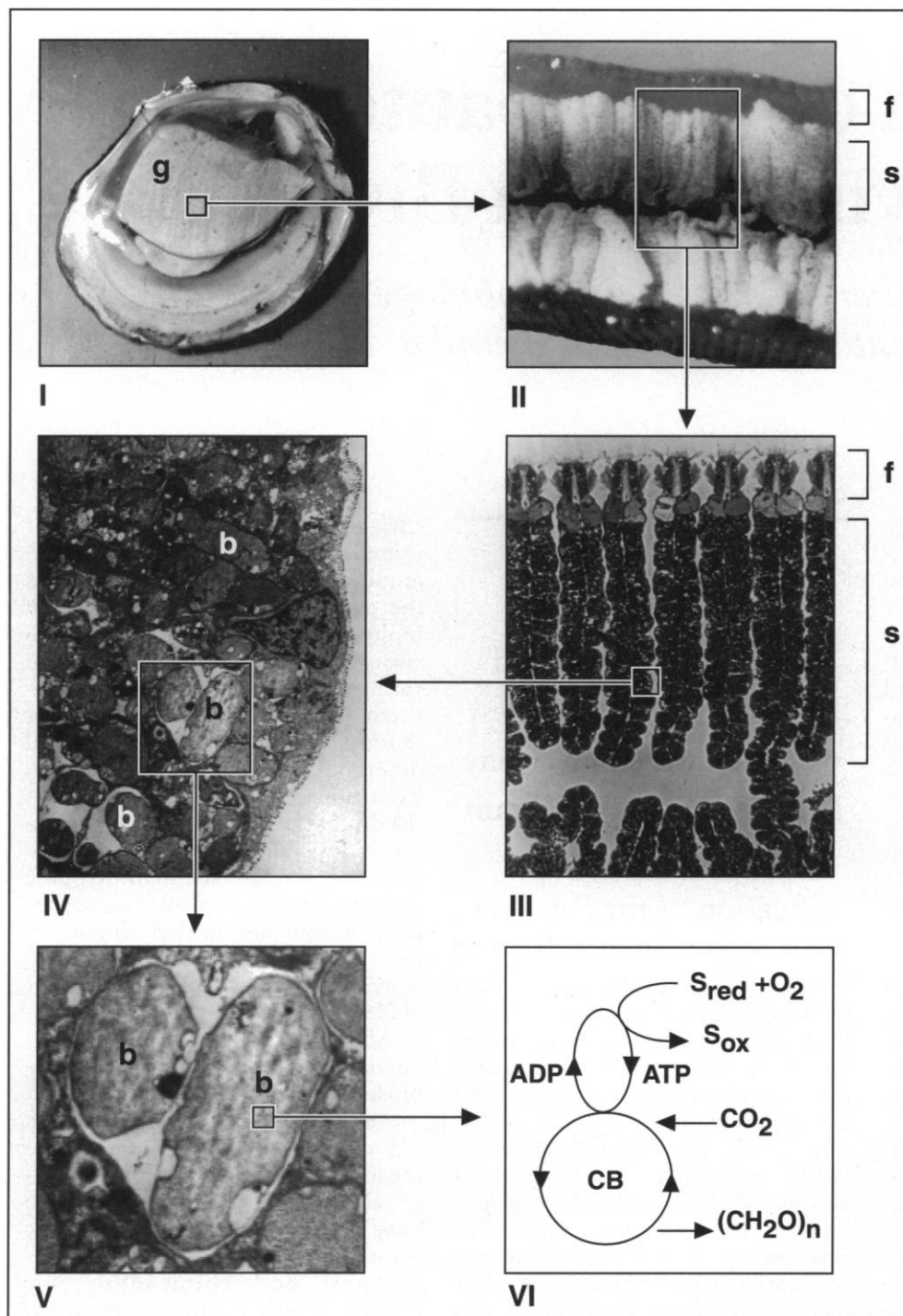


Figure 1. Sequential views of the anatomy of the symbiont-containing gills of *Lucinoma aequizonata* (Lucinidae). I. *L. aequizonata* with the left shell removed to reveal the left demibranch of the paired gills (g). II. A plug of tissue removed from the boxed region in I is shown rotated 90° about its horizontal axis, exposing the transparent filaments (f) and symbiont-containing subfilamentar tissue (s). Note the color difference caused by white bacterial symbionts in the subfilamentar tissue. III. A light micrograph of a stained thick section of tissue from the boxed region in II, showing individual filaments and subfilamentar tissue. IV. An electron micrograph of the boxed region in III shows bacterial symbionts (b) within bacteriocytes in the subfilamentar tissue. V. Detail from boxed region in IV showing two bacterial symbionts. VI. Schematic diagram of sulfur-oxidizing chemoautotrophy in *L. aequizonata* symbionts, which occurs within the symbiont cells shown in the boxed region in V. Oxidation of reduced sulfur compounds (S_{red}) provides energy for production of ATP, which in turn drives fixation of CO_2 into carbohydrate $(CH_2O)_n$ via the Calvin-Benson cycle (CB). This carbohydrate serves as the primary carbon and energy source for the host's nutrition.

their symbiont-bearing organs and their digestive systems.

The symbiont-bearing organs are typically the largest and most conspicuous organs in these animals, often accounting for more than one-third of the animal's total soft-tissue weight. In *Riftia* and other vestimentiferans, this organ (called the trophosome) fills most of the body cavity. Bivalves lack a trophosome; instead, they house their symbionts in specialized cells within the subfilamentar tissue of their gills (Figure 1). Extensive development of this subfilamentar tissue in chemosymbiotic bivalves gives these gills a characteristic thickness and opacity that contrasts sharply with the thin, delicate, translucent gills of most bivalves.

Whereas the symbiont-bearing organs of chemosymbiotic animals are enlarged, their adult digestive systems are either completely absent or are considerably reduced in relative size and complexity by compari-

sons. Chemosymbiotic invertebrates are, thus, unique among higher eukaryotes; their ability to use hydrogen sulfide and carbon dioxide as primary nutrients is as different from "typical" animal heterotrophy as is a plant's ability to use carbon dioxide and light.

The unusual nutritional strategy used by these chemosymbiotic animals is reflected in their unusual appearance and morphology. Vent animals that associate with chemoau-

totrophic symbionts include giant vestimentiferan tube worms (e.g., *Riftia pachyptila*), which may grow to more than two meters in length, and large clams (e.g., *Calyptogena magnifica*) and mussels (e.g., *Bathymodiolus thermophilus*), which may reach nearly one-third of a meter. However, it is not their unusual size that is the most conspicuous difference between these animals and their more typical heterotrophic counterparts, but rather the relative sizes of

son to those of symbiont-free taxa. This observation alone suggests a strong nutritional dependence on the symbionts. In fact, this nutritional relationship has been confirmed for many taxa by extensive physiological, biochemical, ecological, morphological, isotopic, enzymatic, microscopic, and molecular evidence (for reviews, see Felbeck and Somero 1982, Felbeck et al. 1983, 1985, Cavanaugh 1985, 1994, E. Southward 1987, A. Southward 1989, Fisher 1990, Felbeck and Distel 1992).

Although first discovered at hydrothermal vents, chemoautotrophic endosymbioses are not limited to these highly specialized environments. Similar symbiotic mechanisms have been identified in at least five common and widely distributed bivalve families that have long been known to science: the Mytilidae, Solemyidae, Vesicomyidae, Thyasiridae, and Lucinidae (Figure 2). Until recently, however, the nutritional strategies of these symbiont-containing taxa were either unknown or misunderstood. For example, the gutless solemyids were proposed to digest food particles externally, by secreting digestive enzymes into the mantle cavity (Owen 1961), or to absorb dissolved nutrients directly from sea water (Reid 1980, Reid and Bernard 1980), and the lucinids were thought to be adapted for suspension feeding on large particles (Bretsky 1976). These adaptations were proposed to allow lucinids and solemyids to survive in marginal environments unsuitable for most other macrofaunal species (Bretsky 1976).

The environments that these taxa inhabit are also sites of hydrogen sulfide accumulation, even though they are not associated with vents. The sulfides in vent communities come from fluids transported to the sea floor by geothermal flow (hydrothermal vents) or by other hydrological processes (cold seeps); however, in many marine environments, sulfide is produced in situ by biological processes. This biogenic sulfide, produced primarily by bacterial sulfate reduction in anaerobic sediments that are rich in organic material, can also support chemosymbiotic communities. Such communities have been found in mangrove swamps, in sea-grass beds, in anoxic marine ba-

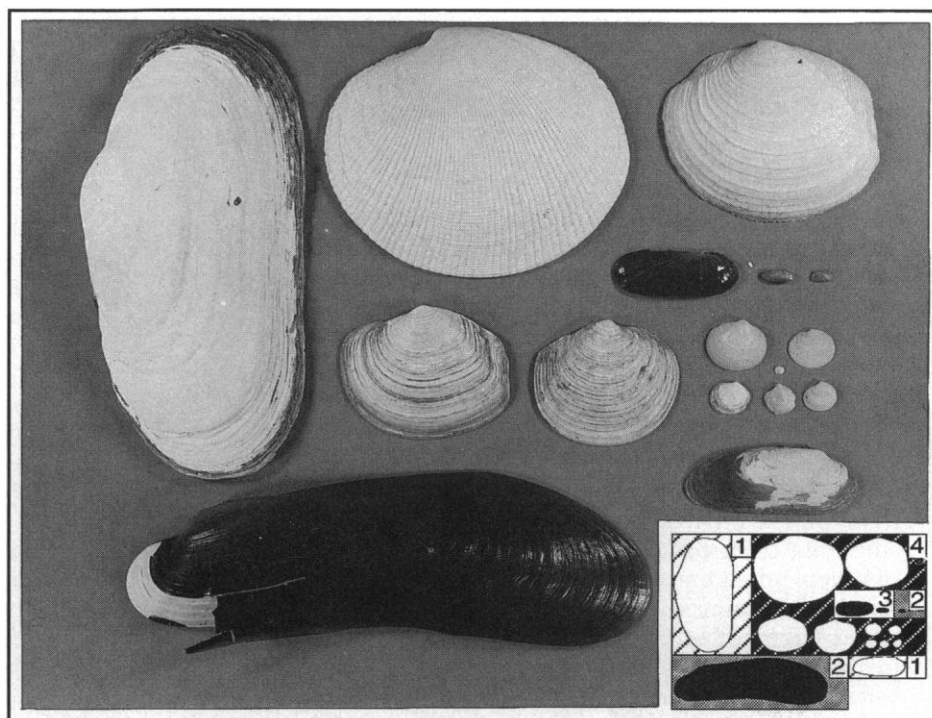


Figure 2. Chemosymbiotic bivalves. Shells from 17 species demonstrate the diversity of chemosymbiotic bivalves. Shaded boxes in the key indicate members of the families Vesicomyidae (1), Mytilidae (2), Solemyidae (3), and the superfamily Lucinacea (4).

sins, in sewage outfalls (Felbeck and Distel 1992), in rotting whale carcasses (Smith et al. 1989), and even in the decaying cargo of a sunken coffee freighter (Dando et al. 1992). The "marginal" nature (i.e., the low macrofaunal diversity) of these environments is likely explained by the toxicity of hydrogen sulfide (a potent inhibitor of electron transport) and the low oxygen concentrations associated with sulfidic sediments (hydrogen sulfide oxidizes spontaneously, so it accumulates only in sediments in which oxygen concentrations are low). Thus, chemoautotrophic symbioses allow their hosts not only to use nutrients normally unavailable to other eukaryotes, but also to inhabit environments that, due to the presence of these same compounds, are inhospitable to most other animals.

Antiquity and interdependence

Because chemosymbiosis was associated with hydrothermal vents, it initially appeared to be an ancient nutritional strategy. Vent communities have been proposed as refugia for descendants of extremely ancient

taxa (Newman 1985, Tunnicliffe 1992) because of their purported similarity to primitive Earth environments, their extreme endemism, and their relative isolation from cataclysmic events affecting Earth's surface. Several lines of evidence now provide additional support for this early impression.

One indication of the antiquity of these symbioses is the degree to which the hosts and symbionts have become adapted to accommodate their symbiotic association. As previously discussed, chemosymbiotic hosts display dramatic modifications in their physiology and body plans that are specifically related to their symbioses. Whereas individual morphological or physiological changes may evolve quickly, integrated and coordinated changes in multiple features of both hosts and symbionts likely reflect long-term selection and "fine tuning" of traits favoring symbiosis specifically.

Obligacy—the loss of the capacity of the host or symbiont for independent existence—may also indicate an ancient association. In most bivalve chemosymbioses, the symbionts are thought to provide a sig-

nificant proportion of the host's carbon nutrition (Southward 1987, 1989, Cary et al. 1989, Conway et al. 1989, 1993, Fisher 1990). Plausible alternative feeding modes often do not exist, particularly in the case of gutless species. Furthermore, aposymbiotic (symbiont-free) adult specimens of chemosymbiotic species are not observed in nature. These observations suggest that the symbioses are obligate for their hosts.

Whether the bacteria found in these hosts are obligate symbionts is, however, uncertain. Although symbionts have neither been successfully cultivated *in vitro* nor observed free-living in the environment, it is possible that these bacteria are capable of independent existence. Most bacteria (particularly endosymbionts) either have not or cannot be grown in pure culture, and cultivation efforts typically underrepresent bacterial diversity in natural environments (Giovannoni et al. 1995).

Some indication of the potential ability of certain symbionts to live independently comes from studies of symbiont transfer mechanisms. Although the precise mechanisms of symbiont transfer are not fully understood in any bivalve system, two distinct modes of symbiont transfer have been proposed. In members of two families (Vesicomyidae and Solemyidae), indirect evidence suggests that symbionts may be present in reproductive tissues, indicating that symbionts may be transferred by a "vertical" mechanism—that is, directly from parent to offspring through the gametes (Cary and Giovannoni 1993, Cary 1994, Krueger et al. 1996). In this case, there would be no need for symbionts to exist apart from their hosts. However, experiments with one lucinid species suggest that its symbionts are acquired from the environment (Gros et al. 1996); if so, these particular symbionts must be capable of free-living existence.

Specificity of bivalve chemosymbioses

Regardless of the mode of transmission, host-symbiont associations are highly species specific, as indicated by comparative sequence analysis of the symbiont's small subunit (16S)

rRNA genes (Distel et al. 1988, 1994, Distel and Wood 1992, Eisen et al. 1992, Distel and Cavanaugh 1994, Kim et al. 1995, Durand et al. 1996, Krueger and Cavanaugh 1997). Each bivalve host harbors just one chemoautotrophic phylotype that is found in all adult specimens, regardless of location or condition. In some cases, the same symbiont phylotype has been observed in more than one host species (Durand et al. 1996). However, this observation does not necessarily indicate that the symbionts are identical or that they may be freely exchanged between these host species. Small subunit rRNA genes evolve at such low rates (Ochman and Wilson 1987, Moran et al. 1993) that organisms with identical 16S rRNA genes may as easily have shared their most recent common ancestor one year or one million years ago.

In a few cases, two symbiont phylotypes have been detected within a single clam (Fisher et al. 1993, Distel and Cavanaugh 1994, 1995). The second phylotype belongs to a phylogenetically and physiologically distinct group of bacteria known as methanotrophs. These symbionts oxidize methane as an energy source and use methane and probably other single-carbon compounds as their primary carbon source. These dual symbioses are unique in that they are the only animal-bacterial symbioses known in which two phylogenetically distinct symbiont phylotypes coexist within a single host cell (Distel and Cavanaugh 1995). Methanotrophic symbioses in bivalves are observed only in the family Mytilidae and can exist exclusive of, or in conjunction with, chemoautotrophic symbioses in different mytilid species (for reviews, see Fisher 1990, Cavanaugh 1993).

In addition to species specificity, chemosymbiotic associations in bivalves exhibit group specificity—that is, closely related bivalve species associate with symbionts that are also closely related (Distel et al. 1994). The occurrence of closely related symbionts in closely related hosts suggests that these particular hosts and symbionts occur together because their most recent common ancestors occurred together in a common ancestral symbiosis—that is,

they are associated by descent (Brooks and McLennan 1993). However, other explanations for group specificity are possible. For example, if symbiotic "infections" are transmitted laterally by colonization of new host taxa, and if these infections are more easily transmitted between closely related hosts than distant ones, then the same pattern of host-symbiont relationships might arise. The distinction between these two scenarios is an important one: If symbionts were acquired before the divergence of their hosts, symbiosis may have been an important evolutionary force driving those divergences, but if symbionts were acquired after host divergence, the symbioses may simply represent similar but independent solutions to a common set of environmental challenges.

History by comparison

It may be possible to distinguish between these alternatives by comparing host and symbiont phylogeny. If symbionts exist primarily within their hosts and are transmitted from generation to generation with extremely high specificity and fidelity, then species divergences among symbionts might eventually be expected to mirror those observed among their hosts. In this case, phylogenetic trees independently inferred for hosts and their respective symbionts should show similar patterns of branch order and branch length. When observed, such congruence is evidence of association by descent and can be used to gauge the antiquity of the associations using the fossil record of the hosts for chronological calibration (Moran and Telang 1998). However, any congruent phylogenetic signals could easily be obscured by sorting events, such as symbiont transfers between host species or *de novo* establishment of symbiotic associations, even if such events are exceedingly rare. Thus, although absolute congruence between host and symbiont trees has a simple interpretation, partial congruence may not.

The reason for this difficulty in interpretation becomes clear when one considers the problems inherent in assessing the evolution of a symbiosis as compared with that of a single organismic lineage. In exam-

ining a symbiosis, at least three histories need to be considered: the history of the hosts, the history of the symbionts, and the history of the association itself. Knowing the history of the symbionts, for example, does not necessarily reveal the history of the symbiosis because hosts may gain or lose symbionts and symbionts may switch hosts. Such reassortment of partners could result in closely related hosts with phylogenetically, or even functionally, unrelated symbionts—that is, host, symbiont, and symbioses may have distinctly different histories. Unless the phylogenetic relationships among hosts and those among symbionts are very nearly parallel, the history of the symbiosis becomes difficult to resolve.

Comparison of host and symbiont phylogenies

Although the phylogeny of chemoautotrophic symbionts has been extensively explored using 16S rRNA analyses, similarly comprehensive molecular analyses of host genes are not yet available. However, comparison of the molecular phylogeny of the symbionts and the classical systematics of the hosts provides insights into the origins and history of these symbioses (Distel et al. 1994).

Chemoautotrophic symbionts examined to date fall into two major lineages within the gamma subdivision of the Proteobacteria (Figure 3). One of these lineages (which I call the Group I chemoautotrophic symbionts) contains symbionts of the bivalve families Solemyidae, Lucinidae, and Thyasiridae, as well as the symbionts of vestimentiferan worms (including *R. pachyptila*; Distel et al. 1988, Feldman et al. 1997) and the ectosymbionts of nematodes (Polz et al. 1994) and oligochaete worms (Dubilier et al. 1995). The second lineage (Group II chemoautotrophic symbionts) contains symbionts of the bivalve families Vesicomidae and Mytilidae. No known free-living or cultivated bacteria fall within either group.

The Group II symbionts can further be subdivided into two distinct monophyletic groups, or clades, one corresponding to the symbionts of mytilids and the second to those of vesicomids. The relationships

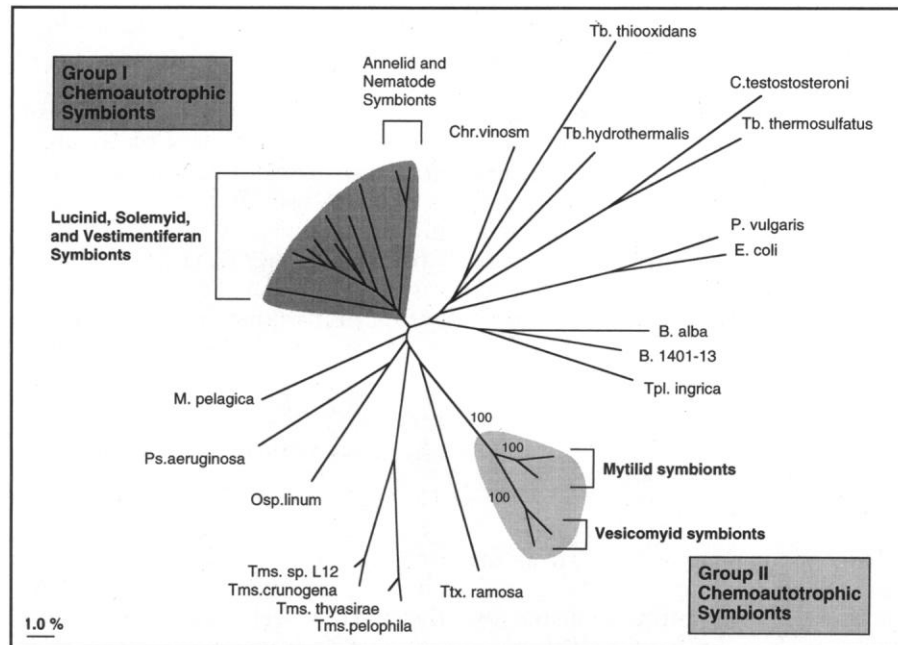


Figure 3. Evolutionary distance tree demonstrating phylogenetic relationships among chemoautotrophic symbionts of bivalves and other marine invertebrates. Reference taxa include chemoautotrophic and heterotrophic representatives of Proteobacteria. Tree was constructed using DNADIST, FITCH, and SEQBOOT with Jukes–Cantor correction (Phylip 3.5; Felsenstein 1989); 1126 nucleotide positions were examined for all taxa. Bootstrap proportions (100 replicates) are shown for selected nodes. B, *Beggiatoa*; C, *Commamonas*; Chr, *Chromatium*; E, *Escherichia*; M, *Methylomonas*; Osp, *Oceanospirillum*; Ps, *Pseudomonas*; Tb, *Thiobacillus*; Tms, *Thiomicrospira*; Tpl, *Thioploca*; Ttx, *Thiothrix*.

among Group I symbionts and their hosts is less clear. Although initial studies showed that known symbionts of lucinaceans (families Lucinidae and Thyasiridae) form a distinct clade (Distel et al. 1994), recent sequence analyses (Krueger 1996) indicate that symbionts of vestimentiferans (a separate phylum) and some solemyids (a separate subclass) also fall within this group. Thus, at least at this level of comparison, Group I symbionts and their host families do not appear to have congruent phylogenies.

These observations suggest that the history of chemoautotrophic endosymbiosis in bivalves is complex and that these associations are likely to have multiple origins. For example, vesicomids belong to the same subclass as lucinids and thyasirids, a subclass distinct from that containing mytilids. The symbionts of vesicomids, however, are closely related to those of mytilids and are distantly related to those of lucinids and thyasirids. These observations are inconsistent with association by de-

scendent. Instead, they suggest that vesicomid symbionts and lucinid/thyasirid symbionts became established in their respective host lineages in separate events, drawing symbionts from bacterial lineages that were already highly divergent.

Similar reasoning suggests that vesicomids and mytilids were already highly divergent families when the vesicomid symbiosis was established. Vesicomidae is thought to be a recent family, with fossil representation only to the Early Cretaceous (95–135 million years before present [mybp]). Interestingly, the appearance of the Vesicomidae coincided roughly with the divergence of vesicomid and mytilid symbionts, as estimated by 16S rRNA divergence rates. A minimum estimated divergence time of 125–300 mybp can be inferred by multiplying the divergences between 16S rRNA sequences available for the two symbiont groups (5–6%) by estimated substitution rates derived from other studies (1–2% divergence per 50 mybp; Ochman and Wilson 1987,

Moran et al. 1993). Could the divergence of vesicomyid and mytilid symbionts reflect the establishment of separate and exclusive associations with Mytilidae and the newly emergent Vesicomyidae? If so, did symbiosis arise independently in these two families? Or was symbiosis first established in one family (perhaps in the more ancient modioloid mytilids) and then transferred to the other? The resolution of these and other questions awaits further phylogenetic analyses.

Diversity of chemoautotrophic symbionts

Why so many diverse hosts are associated with just a few closely related lineages of chemoautotrophic bacteria is not clear. Sulfur-based chemoautotrophy is found in an extraordinarily diverse variety of bacterial lineages, both within the Proteobacteria and in other prokaryotic phyla. There is no *a priori* reason to assume that the physiology of these other bacterial lineages is fundamentally incompatible with chemoautotrophic endosymbiosis. Thus, the particular success of Groups I and II in forming endosymbiotic associations remains unexplained.

Although the reason for their success as chemosymbionts is unknown, the fact that members of Groups I and II have become established as endosymbionts in so many diverse hosts indicates that specialization for association with marine invertebrates has been important to their divergence from other bacterial taxa, much as the specialization for association with a broad range of vertebrate hosts has been an important feature in the evolution of the enteric bacteria. The observation that all known members of Groups I and II are symbiotic suggests that this specialization became established early in their diversification. Like the enterics, Groups I and II may yet be found to contain species that are not symbiotic. Nevertheless, the importance of symbiosis in their evolutionary history is clear.

If symbiotic association is ancestral in Group I chemoautotrophic symbionts, then the diversity of this group indicates that this specialization is extremely ancient—a finding

that agrees well with paleontological data (discussed later). Group I symbionts are highly diverse and are associated with ancient host taxa, whereas the symbionts of Group II are less diverse and are associated with correspondingly more recent host lineages (Distel et al. 1994). This finding suggests that symbioses involving Group II arose separately and perhaps more recently than associations involving Group I.

Diversity of chemosymbiotic hosts

The phylogenetic diversity of chemosymbiotic bivalves also sheds light on the antiquity and origins of chemoautotrophic symbioses. Although relatively few bivalve families host such symbioses, these families are phylogenetically diverse and are distributed among the most distantly related bivalve taxa. Chemosymbiotic lineages have been found in three of the six subclasses of bivalvia (Newell 1969) and are represented among protobranch, filibranch, and eulamellibranch gill types. Paleontological evidence suggests that these three subclasses diverged early in the evolution of the Bivalvia.

Although chemosymbiosis occurs in just a few bivalve families, its pervasiveness within these families suggests that it is evolutionarily important. In four of the five bivalve families that contain chemosymbiotic members, symbionts are present in most, if not all, known species. In the fifth family (the Mytilidae), only a handful of the more than 250 named species associate with chemoautotrophic symbionts (Turner 1985). These symbiont-containing species comprise a single lineage, which was designated as a new subfamily (Bathymodiolinae; Kenk and Wilson 1985). Analyses of 18S rRNA sequences (Daniel L. Distel; Ellen Rice-Kenchington, Department of Fisheries and Oceans, Canada; Eli Chuang, and Colleen M. Cavanaugh, Department of Organismic and Evolutionary Biology, Harvard University; unpublished data) support subfamily status for the Bathymodiolinae, a lineage that, to date, has been found to contain no symbiont-free species. These observations strongly support

the contention that the establishment of symbiosis may have been highly influential in the emergence and subsequent diversification of higher bivalve taxa (Reid and Brand 1986).

Similarities among bivalve chemosymbioses

Despite their considerable phylogenetic diversity, bivalve chemosymbioses exhibit remarkable similarities. In all cases, symbionts are found within the gills, in tissues that are essentially identical in anatomical location and developmental origin. The symbionts of protobranchs, filibranchs, and eulamellibranchs are contained in the intralamellar junctions (Figure 1). In addition, eulamellibranchs also contain symbionts in the interfilamentar junctions. These subfilamentar tissues are far more extensively developed than the homologous tissues in symbiont-free bivalves. In fact, the symbiont-containing subfilamentar tissue typically comprises the bulk of the mass and volume of the gill.

Many similarities are also seen in the cellular architecture of the symbiont-bearing tissues. The symbiont-containing tissues are typically composed of two sheets of simple epithelium surrounding a very narrow central blood sinus. Each epithelium is primarily composed of two cell types in approximately equal proportion. One is the symbiont-containing bacteriocytes, which are roughly cuboidal and which contain basal nuclei and apical symbiont-containing vesicles. The other is the trumpet-shaped, symbiont-free intercalary cells, which expand from a narrow base to a broad, often sheet-like apex that may extend to cover much of the external surface of the neighboring bacteriocytes. Both cell types may contain numerous microvilli but lack cilia and flagella. Large, conspicuous bodies containing numerous concentric membrane whorls are nearly universally observed within bacteriocytes, often in close association with the nuclei. These structures, which are thought to be lysosomal residual bodies, frequently contain objects resembling partially lysed symbiotic bacteria.

The remarkable contrast of morphological similarity and phyloge-

netic diversity in bivalve–bacteria chemosymbioses raises an interesting evolutionary question: Are the similarities due to convergence? Or do the similarities indicate that the presence of gill-borne symbionts is an ancestral condition retained in just a few modern lineages? It can be argued that subfilamentar gill tissue is particularly well suited for providing symbionts with exposure to both the host's blood supply and to dissolved nutrients in seawater, so it is a natural site for symbioses to form convergently. However, other tissues, such as the mantle epithelium or the labial palps, might serve this function equally well. Indeed, endosymbionts have recently been observed in the mantle and foot of one mytilid species (Streams et al. 1997). Even in this species, however, the gills remain the primary symbiont-bearing organ. If, on the other hand, housing bacterial endosymbionts is an ancestral function of the subfilamentar tissues of these diverse chemosymbiotic bivalves, the common symbiont-bearing ancestor would have to be extraordinarily ancient. To predate the divergence of protobranchs, eulamellibranchs, and filibranchs, this hypothetical ancestral host could have arisen no later than the Ordovician and would be counted among the oldest bivalve ancestors.

The fossil record

Some chemosymbiotic bivalve lineages are, in fact, extremely ancient (Figure 4). Lucinaceans are proposed to be the direct descendants of the Babinkacea (McAlester 1966), an extinct superfamily that dates to the Lower Ordovician and contains some of the oldest undisputed bivalve fossils (Cox 1969). Families with chemosymbiotic descendants are also disproportionately well represented among early bivalve fossils. Six recent bivalve families are represented by fossils in Silurian deposits (Kriz 1984); two of these (Lucinidae and Solemyidae) host chemoautotrophic symbionts in all extant species. Although Thyasiridae does not appear until the Cretaceous, this family is thought to have arisen directly from within the Lucinidae (McAlester 1966, Kauffman 1969). Mytilidae appears in the Devonian, but mytilid taxa

Figure 4. First appearances in the fossil record of bivalve taxa with extant chemosymbiotic descendants.

unequivocally associated with chemosymbiotic communities can be traced only to the Late Jurassic (Campbell and Bottjer 1993). Finally, Vesicomidae is the most recent family, making its first known appearance in Cretaceous rocks (Kanie et al. 1993). Thus, the most recent families with exclusively chemosymbiotic descendants appeared more than 65 million years ago, and the most ancient may approach half a billion years in age!

Unfortunately, whether the ancestors of these modern symbiont-bearing bivalve lineages themselves hosted chemoautotrophic symbioses is not known. Bacterial endosymbionts rarely, if ever, leave a trace of their existence in the fossil record. In a few cases, however, isotopic data, based on depletion of the heavy isotope (^{13}C) in chemoautotrophically fixed carbon, has been used to suggest that chemosymbiosis was in place in some fossil bivalves. In these experiments, examination of carbon isotope ratios in the protein matrix preserved in fossilized shells provided evidence that chemosymbioses existed in lucinid bivalves up to approximately 120,000 years old (CoBabe 1991). However, poor preservation of the organic matrix has, so far, prevented the successful use of this technique with older fossils.

Despite the lack of fossil data, the observation that chemoautotrophic symbionts have been found in all

MYBP	System/Series	First Appearance
5	Miocene	← Vesicomidae
23	Oligocene	
53	Eocene	
65	Paleocene	
135	Cretaceous	
205	Jurassic	← Chemosymbiotic mytilids? ← Bivalves at vents
250	Triassic	
290	Permian	
355	Carboniferous	
410	Devonian	← Mytilidae
438	Silurian	← Solemyidae
	Ordovician	← Lucinidae
510		← Babinkacea
570	Cambrian	← First bivalves?

known members of the families Lucinidae, Solemyidae, and Vesicomidae has often been interpreted as evidence that symbioses are ancestral in these families. However, it is also possible that symbioses became established later in the evolution of these families and spread laterally within, and perhaps among, them. Such pervasive transmission, if it were combined with the extinction of nonsymbiotic taxa, could explain the universal occurrence of symbiosis in these families.

Even though symbionts themselves leave no fossil record, paleontological evidence does support the antiquity of these symbioses. Extant chemosymbiotic organisms and those

found in fossil deposits show similar morphological adaptations, life positions, and characteristics of surrounding sediments. These similarities have been interpreted as evidence that ancestral taxa were also adapted to chemosymbiosis (Seilacher 1990, Liljedahl 1991, Campbell and Bottjer 1993). For example, solemyids, lucinids, and thyasirids form unique burrows whose features are specifically related to their simultaneous need to extract sulfide from deep in anaerobic sediments while obtaining oxygen from the overlying sea water. Studies of trace fossils and life positions indicate that these same adaptations were present in ancient species (Bromley 1996).

Evidence supporting the antiquity of chemosymbiosis can also be found in the faunal composition of many fossil bivalve communities. Low in-faunal species diversity and the consistent co-occurrence and predominance of lucinid, thyasirid, solemyid, mytilid, and vesicomyid clams are highly characteristic of modern chemosymbiotic communities. The same features are also apparent in fossil communities (Bretsky 1976, Hickman 1984, Campbell and Bottjer 1993). For example, lucinids, thyasirids, vesicomyids, and mytilids characteristically dominate modern chemosymbiotic communities associated with decaying whale carcasses. Similar species composition has been demonstrated in association with fossil whale bones from the Oligocene (Goedert et al. 1995). Lucinids, solemyids, vesicomyids, and mytilid fossils dominate the Miocene "calcareous Lucina" deposits, which are scattered across the northern Italian Apennines; these sites are now interpreted as chemosymbiotic cold-seep communities (Taviani 1994). Late Eocene deposits in northern Washington contain vesicomyids, mytilids, and thyasirids (Campbell and Bottjer 1993). Lucinid, thyasirid, mytilid, and solemyid fossils co-occur in Late Jurassic cold-seep communities in California (Campbell and Bottjer 1993), and lucinids and solemyids co-occur in Silurian deposits of Gotland (Liljedahl 1991). These observations indicate that features of communities and species interpreted as adaptations to chemosymbioses in modern taxa can be traced to the

earliest appearance of these taxa in the fossil record.

Vents, seeps, and the origins of bivalve chemosymbiosis

Finally, the question remains as to the role of hydrothermal vents and cold seeps in the evolution of chemosymbiosis in bivalves. The oldest vent-associated macrofossil assemblage can be traced to the Silurian Yaman Kasy deposits in the southern Urals of Russia (Little et al. 1997). Although bivalves and tubeworms dominate many modern vent communities, the most ancient fossil vent communities are dominated by brachiopods, tubeworms, and monoplacophorans. In fact, in a survey of 37 fossiliferous sites interpreted to represent Phanerozoic hydrothermal or cold vents, bivalves do not make their first appearance until the Late Jurassic and do not become a predominant component of the vent fauna until the Cretaceous (Campbell and Bottjer 1995). Therefore, the most ancient bivalve chemosymbioses (those hosted by lucinids, thyasirids, and solemyids—that is, those involving Group I symbionts) may have been in existence for hundreds of millions of years before the appearance of bivalves at hydrothermal vents and cold seeps.

The relationship of mytilid and vesicomyid chemosymbioses (i.e., those involving Group II symbionts) to the vent environment is more ambiguous. These families have been observed primarily at vent and seep sites and have been associated with these environments since their earliest known fossil appearances. However, both are also known in organically enriched environments away from vents and seeps, including decaying whale carcasses (whale-falls) and sunken wood. Fossils associated with these environments include mytilids from fossil whale-fall communities and fossil wood, and vesicomyids from turbidity flow deposits in the Oligocene Makah Formation of western Washington (Goedert and Squires 1993, Goedert et al. 1995). Thus, it is not clear whether chemosymbiosis in vesicomyids and mytilids evolved at vents and seeps or whether these chemosymbiotic taxa were simply highly

successful invaders of these environments.

Conclusions

Much remains to be learned about the history of chemosymbiosis in bivalves, but given the present state of knowledge several things seem clear. First, it appears likely that chemosymbioses in bivalves are extremely ancient and extremely successful, allowing some bivalve families to span nearly half a billion years with apparently little change in life habits or appearance. Second, it appears likely that chemoautotrophic endosymbiotic associations existing today have multiple origins, that at least two similar but distinct bacterial lineages are involved, and that host-switching events are likely to have occurred throughout their history. Third, although bivalve chemosymbioses were first discovered at hydrothermal vents and seeps, it now appears unlikely that these symbioses originated in these environments. Finally, the extraordinary diversity and proposed antiquity of chemosymbiotic bivalve and bacterial lineages suggest that chemosymbiosis may have played an important role in the early diversification of the Bivalvia. Continued molecular, physiological, and paleontological explorations promise to reveal much about the remarkable history of this intriguing and important phenomenon.

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